Acceptance Semantics for MAVLink 2 Signing: Bounded Symbolic Analysis and Model-Based Conformance Testing of Flight Stacks

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***Abstract*—The MAVLink 2 message-signing specification fixes a frame format and a timestamp rule but delegates the decisive question—*when is an unsigned or unexpected frame accepted?*—to each implementation. The result is a family of subtly different receivers whose security cannot be compared by reading the specification. We give a precise, parameterised *acceptance semantics* for MAVLink 2 signing as a labelled transition system, in which the choices the specification leaves open become explicit policy parameters, and we equip it with a bounded symbolic adversary and six security properties with machine-checkable definitions. Bounded breadth-first exploration is *sound* (every reported violation is a concrete trace) and *complete up to a depth bound relative to the adversary's restricted move set*; it separates an idealised out-of-band provisioning policy, which satisfies every property while remaining functional, from the policies PX4 and ArduPilot implement, each of which violates a distinct subset. We then use the model generatively: a design-space search over single parameter edits, constrained to preserve a liveness (functionality) property, yields greedy-minimal remediations—and shows, for example, that the current ArduPilot policy requires *two* independent changes where PX4 v1.18 requires one. Finally, we introduce *model-based conformance testing* for flight stacks: read-only test frames derived from the semantics are executed against real software-in-the-loop builds, and the observed accept/reject decisions are compared with the model's predictions. In a recorded experiment across three stacks and four link configurations the model agrees with all 47 observed decisions, and one configuration resolves a disagreement between a stack's documentation and its behaviour in favour of the behaviour; we are explicit that the 20 unkeyed cases are tautological and that the signing-less PX4 v1.17 stack serves as a negative control, so the discriminating evidence rests on the 23 keyed cases, including seven observed rejections. One pair of configurations is a controlled experiment rather than a coincidence of agreement: moving the *same* transport between channel indices flips the decision exactly as the policy's trusted-link parameter predicts. All models, proofs-in-code, and harnesses are released.**

***Index Terms*—formal methods, protocol security, transition systems, bounded model checking, conformance testing, MAVLink, UAV security.**

I. Introduction

A security protocol whose acceptance rule is "left to the implementation" is really a *family* of protocols. MAVLink 2 message signing is such a case: the specification pins the signature construction and a one-minute replay window but states that acceptance of unsigned traffic "depends on the implementation," and offers only informal suggestions (accept `RADIO_STATUS`; an "unsigned mode"; "accept unsigned until the first signed packet"). Two widely deployed receivers, PX4 and ArduPilot, resolve these degrees of freedom differently, and companion measurement work [1] shows the differences have real consequences, including a link that signs every outbound frame yet still accepts unsigned commands. What has been missing is a *model*: a precise object in which the choices are explicit, the adversary is fixed, and the properties are stated formally, so that implementations can be compared, attacked, and repaired by construction.

Contributions.

- A parameterised acceptance semantics for MAVLink 2 signing (Section II): a transition system whose policy parameters capture exactly the specification's open choices, including out-of-band vs. over-the-wire key provisioning.
- A bounded symbolic adversary and six properties (Section III) with a soundness result and depth-bounded completeness (Section IV); exploration separates an idealised policy from the implemented ones.
- A generative use of the model (Section V): a liveness-constrained design-space search that computes *greedy-minimal* remediations per policy (reporting ties as alternatives), yielding non-obvious results (ArduPilot needs two independent fixes; PX4 v1.18 one).
- Model-based conformance testing for flight stacks (Section VI): read-only tests derived from the semantics, executed against real SITL builds as a recorded experiment (47 decisions, three stacks and four link configurations), checking the policies against the running code. One pair of configurations varies a single parameter—which channel index carries a given transport—and the stack's decision flips as the policy predicts, which separates a model of the mechanism from mere agreement.

II. Acceptance Semantics

Definition 1 (Configuration)**.** *A configuration is* $\sigma = (K, armed, S, \mathrm{clk})$ *with key* $K \in \mathcal{K} \cup \{\bot\}$*, arming flag* $armed$*, a partial map* S *from streams* $s = (sys, comp, id)$ *to timestamps, and a local clock* $\mathrm{clk} \in \mathbb{N}$.

Definition 2 (Frame)**.** *A frame is* $f = (\ell, src, \kappa, b, k, t, id)$ *with link* $\ell \in \mathcal{L}$*, source* src*, kind* $\kappa \in \{\textsc{cmd}, \textsc{read}, \textsc{hb}, \textsc{radio}, \textsc{setup}\}$*, signed bit* $b \in \mathbb{B}$*, signing key* k*, timestamp* t*, link id* id*, and (for* SETUP*) a*

TABLE I
POLICIES (ABRIDGED; SEE [1] FOR THE FULL TUPLE).

	trusted	TOFU	oob	u-rekey	allow
spec-strict	∅	yes	no	no	{radio}
ideal-oob	∅	no	yes	no	{radio}
px4-1.18	∅	yes	no	no	{hb,radio}
px4-1.17	all	no	no	—	—
ardupilot-4.7.2	{ch0}	yes	no	yes	{radio}

payload key k'. Under the standard symbolic abstraction, f verifies *in σ iff b and $k = K$: signatures are unforgeable and collision-free.*

Definition 3 (Policy)**.** *A policy π is a tuple* $\langle$trusted $\subseteq \mathcal{L}$, allow $\subseteq$ *Kinds*, openUnkeyed, tofu, oob, uRekey, sRekey, refuseArmed, replay$\rangle$ *of the specification's open choices.*

The transition relation $\sigma \xrightarrow{f/d}_{\pi} \sigma'$, with $d \in \{\mathsf{acc}, \mathsf{rej}\}$, is defined by Algorithm 1 of [1] extended with the out-of-band case: when oob holds the key is factory-installed, so the initial configuration has $K \neq \bot$ and no SETUP frame is ever needed. Freshness is $\mathrm{fresh}(\sigma, f) \equiv \neg\mathsf{replay} \vee (S(s)\uparrow \wedge t + W \geq \mathrm{clk}) \vee (S(s)\downarrow \wedge t > S(s))$ with $W = 6{\times}10^6$.

We instantiate five policies (Table I): spec-strict (strictest reading of the text), ideal-oob (out-of-band keys, no TOFU), and the three measured policies px4-1.17, px4-1.18, ardupilot-4.7.2.

III. ADVERSARY AND PROPERTIES

Definition 4 (Adversary)**.** *The adversary A controls one link a. Its one-step moves are: inject any unsigned frame on a; replay any frame previously observed on a; and inject a frame signed with a key $k \in known(A)$. A cannot forge. Keys enter $known(A)$ exactly when A observes a* SETUP *frame on a (*SETUP *carries k' in clear).*

Properties are predicates over reachable traces. P1 command-integrity: no accepted CMD originates from A on a link $\notin$ trusted. P2 key-integrity: no reachable configuration has $K = K_{\mathrm{adv}}$. P3 replay-resistance: no previously accepted CMD is accepted again. P4 first-use window: $\exists$ a pre-provisioning trace installing K_{adv}. P5 trusted-link exemption: $\exists$ an accepted A-CMD on a link $\in$ trusted. P6 key exposure: with A on the provisioning link, $\exists$ an accepted A-CMD after key leak. P1–P3 are safety failures; P4–P6 are exposure facts. Liveness L: a legitimately provisioned vehicle accepts a legitimate signed CMD on an ordinary link.

IV. BOUNDED SYMBOLIC ANALYSIS

We explore, per policy and per scenario, all adversary action sequences up to depth k by breadth-first search, recording the shortest witness for each property.

Proposition 1 (Soundness)**.** *Every witness returned is a valid trace of $\rightarrow_{\pi}$: the reported violation is realisable by the stated adversary.*

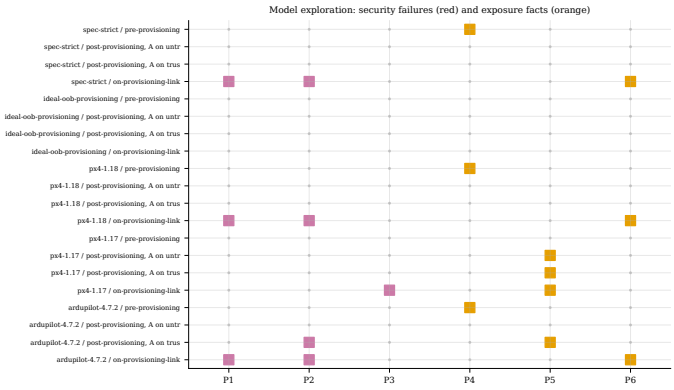


Fig. 1. Exploration (k=3). Red: safety failure witnessed; orange: exposure fact; dot: none. ideal-oob is clean throughout.

Proof sketch. Each search edge applies step_{π} to a reachable configuration and an adversary-constructible frame; the witness is the concatenation of these edges, hence a trace. A-constructibility is enforced by the move generator (unsigned always; signed only with $k \in known(A)$; replay only of observed frames). □

Proposition 2 (Depth-bounded completeness, relative to the move set)**.** *Let M be the adversary's move set actually generated by the exploration: frames with a fixed adversary source identity, a fixed link identifier, and one canonical fresh timestamp per step (the adversary does not spoof the GCS system id, vary the link id, or choose timestamps inside the acceptance window). If a property is violable within k adversary steps* drawn from *M from a scenario's initial configuration (pre-, on-, or post-provisioning, as defined for that property), the search finds a witness of length $\leq k$.*

Proof sketch. For each scenario the search starts from that scenario's initial configuration; BFS over the finitely-branching move relation enumerates all adversary sequences of length $\leq k$ over M. The abstraction makes the per-step frame set finite (bounded kinds, links, keys, and a canonical fresh timestamp), so no violating sequence of length $\leq k$ over M is missed. The proposition says nothing about attacks that need moves outside M (e.g. source-id spoofing or link-id variation against per-stream replay state); those are future work. □

Two caveats keep this claim honest. First, because the abstract clock advances on every step, no two explored configurations coincide, so the visited-set in the implementation never prunes: the search is a bounded tree enumeration, not a closed state-space exploration. Second, the observation that $k = 3$ suffices for the measured policies is *empirical*—no new property becomes violable at $k = 4$ for these policies—and is not a proven saturation. Fig. 1 summarises the outcome. ideal-oob is the unique policy with no witnessed property in any scenario while satisfying L; each implemented policy violates a distinct subset, matching the measurements of [1].

TABLE II
GREEDY-MINIMAL LIVENESS-PRESERVING REMEDIATION (MODEL, k=3).

policy	greedy-minimal edit set clearing P1–P3	L?
spec-strict	close TOFU + out-of-band keys	yes
px4-1.18	close TOFU + out-of-band keys	yes
ardupilot-4.7.2	close TOFU + OOB keys *and* (drop the console-link exemption *or* forbid unsigned re-keying; either clears the failures)	yes
px4-1.17	no single edit reduces the failures: adopt the full mechanism	—
ardupilot-4.7.2-doc	no single edit reduces the failures: the MAVLink 1 downgrade keeps a forged-command path open, so closing the provisioning window no longer suffices	yes

V. GENERATIVE USE: MINIMAL REMEDIATION

Because policies are first-class, we search the design space. For a base policy we consider the single-parameter edits in Table II, recompute the witnessed set, and retain an edit only if it preserves liveness L. A greedy search then builds an edit set clearing all safety failures: an edit is added only if it strictly reduces the failure count, edits that leave a policy unchanged are never counted, iteration order is fixed (so the result is deterministic), and when several edits are equally good at a step all of them are reported as alternatives. The result is *greedy-minimal*; we do not claim global minimality.

Two results are worth stating. First, closing the trust-on-first-use window is only sound *together with* out-of-band provisioning: closing TOFU alone makes the vehicle unprovisionable (it fails L), a trap the liveness constraint catches automatically. Second, ardupilot-4.7.2 requires *two* independent edits—closing the key-exposure window and removing the unsigned re-key on the console link—because the two support distinct P2 witnesses; px4-1.18 requires only the first. px4-1.17, which has no signing implementation, admits no small liveness-preserving fix: the whole mechanism must be added. These are precisely the actionable recommendations a vendor needs, derived rather than asserted.

VI. MODEL-BASED CONFORMANCE TESTING

A policy in Table I is a *hypothesis* about an implementation. We test it. From the semantics we derive read-only probe frames—a `PARAM_REQUEST_READ` (kind READ) signed or unsigned, on a trusted and an untrusted link, in the unkeyed and keyed states—and the reversible key-management steps. For a running SITL build we execute each probe, observe whether the vehicle responds, and compare with step_π's prediction (Algorithm 1). No state-changing command is ever sent.

Agreement turns a documentation-derived policy into a *measured* one; a disagreement is a finding about the model or the implementation. Table III reports the agreement over two campaigns: three stacks in their default link layout (eight decisions each) and four link configurations of ArduPilot 4.7.2 in which we vary the transport of each channel. The model agrees with all 47 recorded decisions. That headline must be read with its evidential weight. Twenty *unkeyed* cases are tautological: with no key installed nothing can reject, so model and stack trivially agree. The four keyed px4-1.17 cases are a *negative control*: the policy encodes "signing absent" and the stack has no signing, so agreement confirms the absence of a mechanism, not a model of one. The discriminating evidence is the remaining 23 keyed cases, and in particular the seven observed *rejections* (px4-1.18: unsigned on the trusted and on the untrusted link; ardupilot-4.7.2: unsigned on the untrusted link in each of the four configurations).

Algorithm 1 Conformance of policy π to a live stack

1: $A \leftarrow 0,\ T \leftarrow 0$
2: **for all** $(\text{keyed}, \ell, b) \in \{0,1\} \times \{\text{trust}, \text{untrust}\} \times \{0,1\}$ **do**
3: ensure vehicle in state keyed (provision once over trusted link)
4: $o \leftarrow$ send unsigned/signed READ on ℓ; observe reply
5: $p \leftarrow \mathrm{step}_\pi(\sigma_{\text{keyed},\ell}, \text{READ}_b).d$
6: $A \leftarrow A + [o{=}p];\ T \leftarrow T + 1$
7: **return** agreement A/T and the disagreeing cases

A controlled pair, not merely agreement: Agreement alone cannot distinguish a model that captures the mechanism from one that happens to predict the same outcome. Configurations B and C differ in exactly one respect: which channel index carries the UDP link. In B the UDP link is `ch0` and a keyed unsigned read is answered; in C the UDP link is `ch1` and the same read is rejected. The policy parameter under test—the set of trusted links, here $\{\texttt{ch0}\}$—is the only thing that can explain the flip, and it does so in both directions. This is the strongest conformance evidence in the paper: the implementation's exemption is keyed to the channel *index*, as the model says, and not to the link's transport, as its source comment's rationale ("USB on ChibiOS boards") would suggest.

A disagreement between documentation and behaviour, resolved: Configuration D sets a port to MAVLink 1, whose frames carry no signature field at all. ArduPilot's signing documentation states that such a port "is not protected", which, read as a policy, says a keyed non-trusted port accepts MAVLink 1 commands. The measurement says otherwise: the keyed port rejects an unsigned MAVLink 1 read, and the probe's frames are recorded as `0xFE`, so they were genuinely version 1 and not unsigned version 2 frames.[1]

We therefore added the wire version to the semantics as a frame attribute ($\mathit{wire} \in \{1,2\}$, with $\mathit{wire}{=}1$ structurally implying unsigned) and the documented behaviour as a policy

[1]We record the first byte actually sent for every case, because a client library may silently upgrade a connection to MAVLink 2 on receiving a version 2 frame; an earlier revision of our harness labelled the wire version from the configuration rather than from the frame, and one attempted measurement against PX4 had to be discarded for exactly this reason.

TABLE III
MODEL-VS-IMPLEMENTATION CONFORMANCE (READ-ONLY PROBES). UPPER BLOCK: DEFAULT LINK LAYOUT. LOWER BLOCK: ARDUPILOT 4.7.2 WITH THE TRANSPORT OF EACH CHANNEL VARIED; CH0 IS THE CHANNEL THE IMPLEMENTATION EXEMPTS.

stack / configuration	policy	agree	keyed rej.
ArduPilot 4.7.2 (default)	ardupilot-4.7.2	8/8	1
PX4 1.18.0-rc1	px4-1.18	8/8	2
PX4 1.17.0 (neg. control)	px4-1.17	8/8	0
A: ch0 TCP, ch1 TCP	ardupilot-4.7.2	6/6	1
B: ch0 **UDP**, ch1 TCP	ardupilot-4.7.2	6/6	1
C: ch0 TCP, ch1 **UDP**	ardupilot-4.7.2	6/6	1
D: ch1 MAVLink 1	ardupilot-4.7.2	5/5	1
total		**47/47**	7

parameter (*v1_bypasses_signing*), and kept the documentation's reading as a separate policy, ardupilot-4.7.2-doc. The parameter earns its place in three ways. It makes the case agree *for the right reason* rather than by the channel rule alone. It adds an adversary move—inject a version 1 frame—that separates the two policies: under the implementation's policy the move is rejected on an ordinary link, under the documentation's reading it delivers a command. And it changes the repair cost: for ardupilot-4.7.2, closing the provisioning window (close-tofu+oob) removes P1, while for the documented reading the same edit does not, because the downgrade keeps a forged-command path open, so the greedy search finds no single sufficient edit at all (Table II).

What the parameter does *not* do is change which properties ArduPilot violates: the channel-0 exemption already violates P1 and P2 on its own, so both readings fail the same set. The difference is in the witnesses and in the cost of repair. We state this plainly because a parameter that changed nothing would still be worth reporting, and this one changes less than its security significance might suggest.

The experiment is *recorded*: the SITL runs are not re-executed in continuous integration; the repository stores the observed decisions and a test asserts that the model reproduces them.

VII. DISCUSSION AND RELATED WORK

Our analysis is a bounded, special-purpose model checker rather than a general-purpose symbolic prover. A ProVerif or Tamarin encoding of the signing and provisioning sub-protocol—capturing unbounded sessions and the truncated-MAC abstraction more faithfully—is natural future work; we provide the semantics in a form amenable to such an encoding. The value of the lightweight model is that it is executable, testable against implementations, and generative. Based on their published abstracts, the only prior formal work on MAVLink models the protocol *without* its native signing and verifies an external cipher [2], [3], or uses session types and runtime monitors with no cryptography [4], [5]; none models the signing scheme itself—its truncated MAC, per-stream timestamp window, and SETUP_SIGNING provisioning. Our methodological lineage is the symbolic analysis of adjacent provisioning protocols: Tamarin for ASTM F3411 Remote ID [6], ProVerif for Bluetooth "Just Works"/passkey [7], [8] and Matter commissioning [9], and Tamarin for automotive SecOC [10]. The closest prior work is PAVE-MAVLink, which pairs a ProVerif analysis of MAVLink 2 with empirical validation, including attacks on the Damn Vulnerable Drone platform and over-the-air validation on a physical PX4 aircraft [2], [11]; its mitigation is an added ChaCha20 layer. We therefore do not claim to be first to verify MAVLink 2 formally, nor first to check such an analysis against running hardware. What we believe is new here is narrower and we state it in those terms: a model whose parameters *are* the points on which implementations differ, so that PX4 and ArduPilot are instances of one semantics rather than separate studies; the treatment of provisioning—the trust-on-first-use window and the per-link exemptions—as first-class rather than as assumptions; a conformance campaign that systematically compares predicted with observed *accept/reject* decisions and accounts for which of them discriminate, rather than demonstrating that particular attacks succeed; and a remedy posed as repair of the native policy rather than as an added cipher. We state this on the basis of published abstracts and metadata—the MILCOM proceedings are paywalled and we have not read the full texts—and we will revise it if those texts show otherwise. Fuzzing of flight stacks [12], [13] targets input validation, a complementary concern: our model assumes memory safety and studies the acceptance decision.

VIII. CONCLUSION

Making the specification's open choices explicit turns "MAVLink signing" into a comparable, attackable, repairable family of policies. Bounded symbolic analysis separates an idealised policy from the deployed ones with concrete witnesses; a liveness-constrained search derives greedy-minimal fixes (with equally good alternatives made explicit) and reveals that ArduPilot needs two where PX4 v1.18 needs one; and model-based conformance testing, honestly accounted, ties the discriminating part of the construction back to the running code.

AVAILABILITY

The executable semantics, exploration, synthesis, conformance harness, and tests are released at the project repository.[2] The work is defensive and uses only self-hosted simulators.

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